

Daffodil International University
Faculty of Science & Information Technology
Department of Computer Science & Engineering
Final Examination, Spring 2026
Course Code: CSE321, Course Title: System Analysis and Design
Batch: RE_A(3C)

Time: 02:00 Hrs	Marks: 40
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Answer ALL Questions

[The figures in the right margin indicate the full marks and corresponding course outcomes. All portions of each question must be answered sequentially.]

You have been assigned to design an Online University Admission Management System for a university with three campuses located in Dhaka, Rajshahi, and Chattogram. The admission office operates from 9 am to 5 pm and manages admission for multiple departments under faculties such as Science & Engineering, Business, and Arts & Social Science. The system must handle three application categories: Undergraduate (application fee Tk. 600), Graduate (Tk. 800), and PhD (Tk. 1,200) along with a mandatory document verification fee of Tk. 200. The system must support online payment through multiple gateways (bKash, card, and bank transfer). Applicants are required to submit academic transcripts, national ID, and passport-sized photographs. For security, all applicant accounts must follow strong password policies and device-based login with OTP verification for new devices. Applicants can check seat availability across departments in real time, and the system must track application statuses and generate admission result reports. Entry to the exam hall is controlled using a digital admit card with a QR-code, which becomes invalid if application fees are unpaid. Applicants also must fill out a background information form at registration so that the system can recommend suitable programs based on academic history and interests. Your task is to design a system that efficiently manages applications, payments, document verification, scheduling, and secure access control across all campuses while ensuring a smooth user experience. Now answer the following questions.

1.	a.	Explain how the Online Admission System can leverage digital marketing channels (email campaigns, social media ads, SMS alerts, and push notifications) to attract prospective applicants and improve enrollment rates.	[3]	CO1
	b.	Identify the most appropriate e-commerce business model (B2B, B2C, C2C, or C2B) for the Online University Admission System and justify your selection. Also compare three electronic payment systems (bKash, credit/debit card, and bank transfer) and argue which is the most suitable for this university admission system.	[5]	CO1
2.	a.	Design a Monthly Application Summary Invoice for applicants. It should include Invoice Number, Applicant Details, Application Category (UG/Graduate/PhD), Application Fee, Document Verification Fee (if applicable), Optional Services (Hostel Registration, Transport), Discount (if any), Total Payable Amount, Payment Status (Paid/Unpaid), Due Date, and Payment Gateway Reference ID.	[5]	CO2
	b.	Create test cases and test data required to evaluate the functionality of the QR-code-based digital admit card login/access system for the examination hall.	[7]	CO2

	c.	Analyze the system and outline a security and access control strategy that protects applicant information, prevents unauthorized access to the admission portal, and ensures data integrity across all three campuses.	[5]	CO2		
3.	The following table shows the activities, dependencies, and three-time estimates (optimistic, realistic, pessimistic) required to complete the Online Admission Management System. Based on this:		[2] [4] [4] [5]	CO3		
	Activity	Activity Name			Dependency	Duration (o, r, p) weeks
	A	Project Initiation			—	(2, 4, 6)
	B	Requirements Analysis			A	(4, 6, 8)
	C	UI/UX & System Design			—	(1, 3, 5)
	D	Applicant Registration Module			A, C	(3, 5, 7)
	E	Payment & Fee Module			B	(1, 3, 5)
	F	Document Verification Module			C	(2, 4, 6)
	G	Admit Card & QR Module			D, E	(3, 5, 7)
	H	Reporting & Analytics			D, F	(5, 7, 9)
	I	System Integration & Testing			G, H	(3, 5, 7)
	J	Multi-campus Sync & Deployment			E, F	(4, 6, 8)
	K	Go-Live & Handover			I, J	(2, 4, 6)
(i) Apply the time sequence formula to calculate the estimated duration for all activities in the project.						
(ii) Sketch a Gantt Chart for the project timeline using the estimated durations calculated in part (i). Using your calculations, and the total project duration.						
(iii) Show forward and backward calculation and determine the Early Start (ES), Early Finish (EF), Late Start (LS), and Late Finish (LF) for all activities and determine the complete Critical Path.						
(iv) The client unexpectedly requests a change that adds 3 extra weeks to Activity F. Evaluate its impact on the Critical Path, Total project duration, Slack distribution of affected tasks, and whether project delay is avoidable or not. (Show calculation and reasoning.)						